

Sustainable Engineering Design and Evidence Rubric

TEACHER USE

3sw/wk3-sustainable-engineering

Recommended major evidence packet: 16 points

Score the final Pest Patrol design, revision record, and Trends Evaluation together. Paper, Canva, Adobe Express, or another approved design route is equal. Xello completion, drawing skill, platform access, and public speaking are not separate grades.

CRITERION	MASTERS - 4	MEETS - 3	APPROACHES - 2	NEEDS IMPROVEMENT - 1	NO EVIDENCE - 0
Constraint and evidence reasoning	Uses three clear constraints and connects at least three design features to specific field-report evidence	Three constraints and evidence links are present; one link is thin	Two usable constraints or mostly general evidence links	Partial design with little source evidence	Blank or off topic
Functional communication	Six or more readable labels explain sensing, movement, reporting, and safety/crop protection	Six labels are present and most functions are clear	Four or five usable labels; one required function is unclear	A drawing is present but the reader must guess how it works	Blank or unusable
Revision and tradeoff	Makes a visible evidence-based revision and explains a specific benefit plus cost, risk, or limit	Revision is visible and a reasonable tradeoff is named	Revision or tradeoff is present but weakly explained	Feedback is copied or acknowledged without a useful change	No revision evidence
Trend and career evaluation	Compares two trends, uses two accurate sourced facts, explains changed work, and states an evidence limit	Uses two facts and gives a defensible career recommendation	Uses one accurate fact and a general trend connection	Preference is stated with little or inaccurate evidence	Blank or off topic

Score record

CRITERION	POINTS
Constraint and evidence reasoning	/4
Functional communication	/4
Revision and tradeoff	/4
Trend and career evaluation	/4
Total	/16

Performance-band conversion

RAW POINTS	PERCENT RANGE	BAND
15-16	90-100	Masters
13-14	80-89	Meets
12	70-79	Approaches
10-11	60-69	Needs Improvement
0-9	Below 60	Follow campus or district policy

Scoring boundaries

- Score the evidence chain and function, not artistic polish or access to a premium design tool.
- One visible revision is enough when the student can explain why it improves the work.
- Peer feedback is formative. Do not reduce a score because a reviewer missed a problem.
- BLS figures are national medians and projections. Do not reward claims that present them as DFW starting pay or guaranteed outcomes.
- English mechanics do not lower the score unless the meaning cannot be understood.
- Students may revise according to the teacher's normal major-assignment practice.